



Trigonometric Equations Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 The expression $2\cos x - 3\sin x$ is written in the form $R\cos(x + \theta)$, where $R > 0$ and $0 \leq \theta \leq \frac{\pi}{2}$. 1

What is the value of $\tan \theta$?

$$2\cos x - 3\sin x = R\cos(x + \theta)$$

$$= R\cos x \cos \theta - R\sin x \sin \theta$$

Equating both sides:

$$2\cos x = R\cos x \cos \theta \text{ and } 3\sin x = R\sin x \sin \theta$$

Therefore:

$$R\cos \theta = 2 \quad (1)$$

$$R\sin \theta = 3 \quad (2)$$

$$(2)$$

$$(1)$$

$$\therefore \tan \theta = \frac{3}{2}$$

- 2 By expressing $3\cos 3x - 4\sin 3x$ in the form $R\cos(x + \alpha)$, solve $3\cos 3x - 4\sin 3x = 5$, for 3

$$0 \leq x \leq \frac{2\pi}{3}$$

$$R\cos(x + \alpha) = R\cos x \cos \alpha - R\sin x \sin \alpha$$

$$= 3\cos 3x - 4\sin 3x$$

$$R^2 = 3^2 + 4^2 = 5^2 \therefore R = 5$$

$$5\cos \alpha = 3, \quad 5\sin \alpha = 4 \quad \tan^{-1} \frac{4}{3}$$

[α lies in the first quadrant]

$$5\cos(3x + \tan^{-1} \frac{4}{3}) = 5$$

$$\cos(3x + \tan^{-1} \frac{4}{3}) = 1$$

$$3x + \tan^{-1} \frac{4}{3} = 0, 2\pi$$

$$\therefore x = \frac{-\tan^{-1} \frac{4}{3}}{3}, \quad \frac{2\pi - \tan^{-1} \frac{4}{3}}{3}$$

x not a solution

$$\therefore x = \frac{2\pi - \tan^{-1} \frac{4}{3}}{3} \quad \text{or} \quad \frac{2\pi - 51.8}{3} \text{ degrees}$$

- 3 Find the maximum value of $y = \sqrt{3}\cos x + \sin x$. 2

$$\text{Let } \sqrt{3}\cos x + \sin x = R\sin(x + \alpha)$$

Hence there is no vertical shift, and the amplitude will give the maximum value.

$$R^2 = \sqrt{3}^2 + 1^2 = 4$$

So maximum value $= R = 2$.

- 4 Use t - formulae to solve the equation $\cos x - \sin x = 1$, where $0 \leq x \leq 2\pi$. 3

$$\cos x - \sin x = 1$$

$$\text{Let } t = \tan \frac{x}{2} \quad (x = \pi \text{ is not a solution})$$

$$\therefore \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = 1$$

$$1-t^2-2t=1+t^2$$

$$0 = 2t^2 + 2t = 2t(t+1)$$

$$\therefore \tan \frac{x}{2} = t = 0, -1 \quad \frac{x}{2} = 0, \pi, \frac{3\pi}{4} \rightarrow x = 0, 2\pi, \frac{3\pi}{2}$$

- 5 (i). Express $\sqrt{3}\cos x + \sin x$ in the form $R\cos(x - \alpha)$ where $R > 0$ and α is acute. 2

$$\begin{aligned}\sqrt{3}\cos x + \sin x &= R\cos x \cos \alpha + R\sin x \sin \alpha \\ R\cos \alpha &= \sqrt{3} & R\sin \alpha &= 1 & R &= \sqrt{3+1} = 2 \\ \cos \alpha &= \frac{\sqrt{3}}{2} & \sin \alpha &= \frac{1}{2} & \alpha &= \frac{\pi}{6} \\ \tan \alpha &= \frac{1}{\sqrt{3}} \\ \alpha &= \frac{\pi}{6} \\ \therefore \sqrt{3}\cos x + \sin x &= 2\cos\left(x - \frac{\pi}{6}\right)\end{aligned}$$

- (ii). Hence solve $\sqrt{3}\cos x + \sin x = 1$ for $0 \leq x \leq 2\pi$. 2

$$\begin{aligned}(ii) \quad 2\cos\left(x - \frac{\pi}{6}\right) &= 1 & \checkmark \text{ valid } \alpha = \frac{\pi}{6} \\ \cos\left(x - \frac{\pi}{6}\right) &= \frac{1}{2} \\ \therefore x - \frac{\pi}{6} &= \frac{\pi}{3} \quad \text{or} \quad x - \frac{\pi}{6} = \frac{5\pi}{3} \\ x &= \frac{\pi}{3} + \frac{\pi}{6} & x &= \frac{5\pi}{3} + \frac{\pi}{6} \\ &= \frac{\pi}{2} \quad \text{or} & &= \frac{11\pi}{6} \quad [2]\end{aligned}$$

- 6 The ampere (or amp) is a unit used to measure electric current. The current i in amperes, at time t , in a circuit is calculated using the equation

$$i = 12\sin t + 5\cos t$$

- (i). Write an expression for i in the form $R\sin(t + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. Give the value of α in radians correct to two decimal places. 2

$$\begin{aligned}R &= \sqrt{12^2 + 5^2} = 13 \\ \alpha &= \tan^{-1}\left(\frac{5}{12}\right) = 0.39 \text{ (2 decimal places)} \\ \therefore i &= 13\sin(t + 0.39)\end{aligned}$$

- (ii). Using the result in part (i) or otherwise, find the maximum current in the circuit and the first time it occurs. Give your answer to two decimal places. 2

Since the amplitude = 13, the maximum value of $i = 13$.

This occurs when:

$$\sin(t + 0.39) = 1$$

$$t + 0.39 = \frac{\pi}{2}, \frac{5\pi}{2}, \dots$$

$\therefore$ first time maximum current occurs is at $t = 1.18$ seconds (2 decimal places).

- 7 Use t - identities to solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $[0, 2\pi]$. 3

$$\begin{aligned}\frac{\sqrt{3}\sin\theta - \cos\theta}{\sqrt{1+1}} &= \frac{1}{\sqrt{2}} \\ \frac{\sqrt{3} \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}}{2\sqrt{3}t - 1 + t^2} &= \frac{1+t^2}{1+t^2} \\ 2\sqrt{3}t - 1 + t^2 &= 1 + t^2 \\ 2\sqrt{3}t &= 2 \\ t &= \frac{1}{\sqrt{3}} \\ \tan \theta/2 &= \frac{1}{\sqrt{3}} \\ \theta/2 &= \frac{\pi}{6} \quad \text{or} \quad \pi + \frac{\pi}{6} \\ \theta &= \frac{\pi}{3} \quad \text{or} \quad \frac{7\pi}{3} \\ \text{but } 0 \leq \theta &\leq 2\pi \\ \therefore \theta &= \frac{\pi}{3} \\ \text{Also check } x &= \pi, \\ \text{LHS} &= \sqrt{3}\sin\pi - \cos\pi \\ &= 0 - (-1) = 1 = \text{RHS} \\ \therefore \text{Solution is } x &= \frac{\pi}{3} \quad \text{or} \quad \pi\end{aligned}$$

- 8 (i). Express $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$, where α is an acute angle.

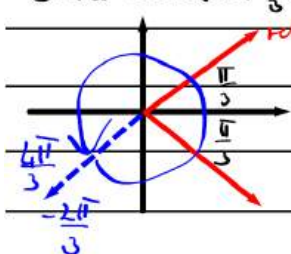
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$$\begin{aligned} \text{i) } \cos x - \sqrt{3}\sin x &= R\cos(x + \alpha) \\ \cos x - \sqrt{3}\sin x &= R\cos x \cos \alpha - R\sin x \sin \alpha \\ \therefore R\cos \alpha &= 1 \quad /^2 \\ + R\sin \alpha &= \sqrt{3} \quad /^2 \\ R^2(\cos^2 \alpha + \sin^2 \alpha) &= 4 \\ R^2 &= 4 \\ \therefore R &= 2 \text{ as } R > 0 \end{aligned}$$

$$\begin{aligned} \text{Since } \cos \alpha &= \frac{1}{2} > 0 \text{ and} \\ \sin \alpha &= \frac{\sqrt{3}}{2} > 0 \\ \alpha &\text{ is in 1st quadr and} \\ \alpha &= \frac{\pi}{3} \\ \therefore \cos x - \sqrt{3}\sin x &= 2\cos(x + \frac{\pi}{3}) \quad [2] \end{aligned}$$

- (ii). Hence, solve $\cos x - \sqrt{3}\sin x = 1$ for $-\pi \leq x \leq \pi$.

2

$$\begin{aligned} \text{ii) } \cos x - \sqrt{3}\sin x &= 1 \quad -\pi \leq x \leq \pi \\ \therefore 2\cos(x + \frac{\pi}{3}) &= 1 \\ \therefore \cos(x + \frac{\pi}{3}) &= \frac{1}{2} \quad -\frac{2\pi}{3} \leq x + \frac{\pi}{3} \leq \frac{4\pi}{3} \\ \text{Since } \cos(x + \frac{\pi}{3}) > 0 &\quad \text{rel } < = \frac{\pi}{3} \\ x + \frac{\pi}{3} &= -\frac{\pi}{3}, \frac{\pi}{3} \\ \therefore x &= -\frac{2\pi}{3}, 0 \quad [2] \end{aligned}$$


- 9 The equation $I(t) = 4\sin 0.02t + 8\cos 0.02t$ measures the current, I amperes, in a particular circuit after t seconds.

- (i). Express the function in the form $I(t) = A\sin(at + b)$, for $0 \leq b \leq \frac{\pi}{2}$.

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$$\begin{aligned} I(t) &= 4\sin 0.02t + 8\cos 0.02t \\ I(t) &= A\sin at \cos b + A\cos at \sin b \\ \text{Equate the expressions} \end{aligned}$$

$$\begin{aligned} \text{since } A > 0 \quad \therefore A &= 4\sqrt{5} \\ \frac{\sin b}{\cos b} &= \frac{8}{4} \\ \tan b &= 2 \\ b &= \tan^{-1}(2) \end{aligned}$$

$$\begin{aligned} a &= 0.02 \quad A\cos b = 4 \quad A\sin b = 8 \\ A^2\cos^2 b + A^2\sin^2 b &= 4^2 + 8^2 \\ A^2(\cos^2 b + \sin^2 b) &= 4^2 + 8^2 \\ A^2(1) &= 4^2 + 8^2 \\ A &= \pm\sqrt{80} \end{aligned}$$

$$\therefore I(t) = 4\sqrt{5} \sin(0.02t + \tan^{-1}(2))$$

- (ii). When does the current first reach its maximum value?

3

A sine graph first reaches its max value at $\frac{\pi}{2}$

$$\text{For } I(t) = 4\sqrt{5} \sin(0.02t + \tan^{-1}(2))$$

$$\begin{aligned} 0.02t + \tan^{-1}(2) &= \frac{\pi}{2} \\ 0.02t &= \frac{\pi}{2} - \tan^{-1}(2) \\ t &= \frac{\frac{\pi}{2} - \tan^{-1}(2)}{0.02} \end{aligned}$$

$t = 23$ seconds, first reaches max at 23 seconds.



Trigonometric Equations Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Given $R\sin(x + \alpha) = \sqrt{3}\sin x - \cos x$, where $R > 0$ and $-\pi \leq \alpha \leq \pi$, find the values of R and α . 1

$$\begin{aligned} R\sin(x+\alpha) &= \sqrt{3}\sin x - \cos x \\ R\sin x \cos \alpha + R\cos x \sin \alpha &= \sqrt{3}\sin x - \cos x \\ \therefore R\cos \alpha &= \sqrt{3} \\ R\sin \alpha &= -1 \end{aligned} \Rightarrow \frac{\sqrt{3}}{R} \quad \frac{-1}{R}$$

$$R=2, \alpha = -\frac{\pi}{6}$$

- 2 (i). Write $3\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$. 2

$$\begin{aligned} 3\cos x - \sqrt{3}\sin x &\equiv R\cos(x+\alpha) \\ &= R\cos x \cos \alpha - R\sin x \sin \alpha \\ \text{Equating co-efficients of like terms} \\ 3 &= R\cos \alpha \dots (1) \text{ and } \sqrt{3} = R\sin \alpha \dots (2) \\ (1)^2 + (2)^2 & \quad 12 = R^2(\cos^2 \alpha + \sin^2 \alpha) \\ 2\sqrt{3} &= R \quad (\text{assuming } R > 0) \\ (2) \div (1) & \quad \tan \alpha = \frac{1}{\sqrt{3}} \\ \alpha &= \pi/6 \quad (\text{assuming } 0 < \alpha < \pi/2) \end{aligned}$$

- (ii). Hence Solve $3\cos x - \sqrt{3}\sin x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. 2

$$\begin{aligned} 3\cos x - \sqrt{3}\sin x &= \sqrt{3} \\ 2\sqrt{3}\cos(x+\pi/6) &= \sqrt{3} \\ \cos(x+\pi/6) &= \frac{1}{2} \\ x+\pi/6 &= \pi/3 \text{ or } 5\pi/3 \\ x &= \pi/6 \text{ or } 3\pi/2 \end{aligned}$$

- 3 (i). Prove the identity $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 2

$$\begin{aligned} \text{LHS} &= \cos 3\theta = \cos(2\theta + \theta) \\ &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta = (2\cos^2 \theta - 1)\cos \theta - 2\sin \theta \cos \theta \sin \theta \\ &= 2\cos^3 \theta - \cos \theta - 2\sin^2 \theta \cos \theta = 2\cos^3 \theta - \cos \theta - 2\cos \theta(1 - \cos^2 \theta) \\ &= 2\cos^3 \theta - \cos \theta - 2\cos \theta + 2\cos^3 \theta = 4\cos^3 \theta - 3\cos \theta \end{aligned}$$

- (ii). Hence, find exact value expressions for three unique solutions of $8x^3 - 6x = -\sqrt{3}$. 3

$$\begin{aligned} 8x^3 - 6x &= -\sqrt{3} \quad (\div 2) \\ 4x^3 - 3x &= -\sqrt{3}/2 \\ \text{Let } x &= \cos \theta, \\ 4\cos^3 \theta - 3\cos \theta &= -\sqrt{3}/2 \\ \cos 3\theta &= -\sqrt{3}/2 \quad \checkmark \text{ for } 0 \leq 3\theta \leq 6\pi \\ 3\theta &= \cos^{-1}(-\sqrt{3}/2) \\ &= \pi - \pi/6, \pi + \pi/6, 3\pi - \pi/6, 3\pi + \pi/6, 5\pi - \pi/6, 5\pi + \pi/6 \\ &= \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{17\pi}{6}, \frac{19\pi}{6}, \frac{29\pi}{6}, \frac{31\pi}{6} \quad \checkmark \\ \theta &= \frac{5\pi}{18}, \frac{7\pi}{18}, \frac{17\pi}{18}, \frac{19\pi}{18}, \frac{29\pi}{18}, \frac{31\pi}{18} \\ &\text{for } 0 \leq \theta \leq 2\pi \end{aligned}$$

$$\begin{aligned} \text{but } x &= \cos \theta \\ x &= \cos \frac{5\pi}{18}, \cos \frac{7\pi}{18}, \cos \frac{17\pi}{18} \quad \checkmark \\ \text{Note } \cos \frac{19\pi}{18} &= \cos(\pi + \pi/18) = \cos(\pi - \pi/18) = \cos \frac{17\pi}{18} \\ &\quad \text{P.T.O.} \\ \cos \frac{29\pi}{18} &= \cos(\pi + \frac{11\pi}{18}) = \cos(\pi - \frac{11\pi}{18}) = \cos \frac{7\pi}{18} \\ \cos \frac{31\pi}{18} &= \cos(\pi + \frac{13\pi}{18}) = \cos(\pi - \frac{13\pi}{18}) = \cos \frac{5\pi}{18} \end{aligned}$$

4 Use sums to products of trigonometric ratios to solve the equation:

3

$$\cos x + \cos 2x + \cos 3x = 0, 0 \leq x \leq 2\pi.$$

$$\begin{aligned} \cos x + \cos 2x + \cos 3x &= 0; \quad 0 \leq x \leq 2\pi \\ \left(\text{Using } \cos A \cos B &= \frac{1}{2} [\cos(A-B) + \cos(A+B)] \right) \\ \cos x + \cos 3x &= 2 \cos 2x \cos x \\ \cos x + \cos 2x + \cos 3x &= 0 \\ \Rightarrow \cos 2x + 2 \cos 2x \cos x &= 0 \\ \cos 2x (1 + 2 \cos x) &= 0 \quad (0 < 2x \leq 2\pi) \\ \cos 2x &= 0 \quad \cos x = -\frac{1}{2} \\ 2x &= \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \quad x = \frac{2\pi}{3}, \frac{4\pi}{3} \\ \therefore x &= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{2\pi}{3}, \frac{4\pi}{3} \end{aligned}$$

5 (i). Express $3\cos x + 2\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$.

3

$$\begin{aligned} \text{i) } 3 \cos x + 2 \sin x &= R \cos(x - \alpha) \\ &= R \cos x \cos \alpha + R \sin x \sin \alpha \\ \text{Equating coefficients, } R \cos \alpha &= 3 \quad \text{--- (1)}; \quad R \sin \alpha = 2 \quad \text{--- (2)} \\ \text{Squaring and adding (1) and (2)} \\ R^2 \cos^2 \alpha + R^2 \sin^2 \alpha &= 9 + 4 \\ R^2 (\cos^2 \alpha + \sin^2 \alpha) &= 13 \\ R &= \sqrt{13}, \quad R > 0 \\ \cos \alpha &= \frac{3}{\sqrt{13}}, \quad \sin \alpha = \frac{2}{\sqrt{13}} \quad \therefore 0^\circ < \alpha < 90^\circ \\ \alpha &= 33.69^\circ \text{ (to 2dp)} \\ \therefore 3 \cos x + 2 \sin x &= \sqrt{13} \cos(x - 33.69^\circ) \end{aligned}$$

(ii). Hence, solve $3\cos x + 2\sin x = \sqrt{13}$, $0^\circ \leq x \leq 360^\circ$. Given your answer in degrees

2

correct to two decimal places.

$$\begin{aligned} 3 \cos x + 2 \sin x &= \sqrt{13} \\ \Rightarrow \sqrt{13} \cos(x - 33.69^\circ) &= \sqrt{13} \\ \cos(x - 33.69^\circ) &= 1 \\ -33.69^\circ \leq x - 33.69^\circ &\leq 326.31^\circ \\ \therefore x - 33.69^\circ &= 0^\circ \\ \therefore x &= 33.69^\circ \end{aligned}$$

6 By expressing $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$ where $R > 0$, solve the equation

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$$\sqrt{3}\sin\theta - \cos\theta - 1 = 0 \text{ for } [0, 2\pi].$$

$$\begin{aligned} \sqrt{3} \sin \theta - \cos \theta &= R \sin(\theta - \alpha) \\ \sqrt{3} \sin \theta - \cos \theta &= R \sin \theta \cos \alpha - R \cos \theta \sin \alpha \\ \text{Equating parts} \\ R \cos \alpha &= \sqrt{3} \quad \text{--- (1)} \quad \text{and} \quad R \sin \alpha = 1 \quad \text{--- (2)} \\ \text{(1)}^2 + \text{(2)}^2 \quad R^2 \sin^2 \alpha + R^2 \cos^2 \alpha &= 1 + 3 \\ R^2 (\sin^2 \alpha + \cos^2 \alpha) &= 4 \\ R^2 &= 4 \\ R &= 2, \quad R > 0 \\ \cos \alpha &= \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin \alpha = \frac{1}{2} \\ \therefore \alpha &\text{ is in 1st Quad} \\ \therefore \sin \alpha &= \frac{1}{2} \\ \cos \alpha &= \frac{\sqrt{3}}{2} \\ \tan \alpha &= \frac{1}{\sqrt{3}} \quad \therefore \alpha = \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} \therefore \sqrt{3} \sin \theta - \cos \theta &= 2 \sin(\theta - \frac{\pi}{6}) \\ \text{Now} \\ \sqrt{3} \sin \theta - \cos \theta &= 1 \\ \text{i.e. } 2 \sin(\theta - \frac{\pi}{6}) &= 1 \\ \sin(\theta - \frac{\pi}{6}) &= \frac{1}{2} \quad \text{if } 0 \leq \theta \leq 2\pi \\ \therefore \theta - \frac{\pi}{6} &= \frac{\pi}{6}, \frac{5\pi}{6} \\ \text{and} \\ \theta &= \frac{\pi}{3}, \pi \end{aligned}$$

7 Using t - formulae, solve the equation $\sin 2x = \tan x$ for $0 \leq x \leq \pi$.

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let $t = \tan x$
 then $\sin 2x = \frac{2t}{1+t^2}$ check where $\tan x$ is undefined
 $\frac{2t}{1+t^2} = t$
 $2t = t + t^3$
 $t^3 - t = 0$
 $t(t^2 - 1) = 0$
 $t(t-1)(t+1) = 0$
 $t = 0, 1 \text{ or } -1$
 $\tan x = 0, \tan x = 1, \tan x = -1$
 $x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \text{ or } \frac{5\pi}{4}$

8 (i). Show that $2\cos(x + \frac{\pi}{3}) = \cos x - \sqrt{3} \sin x$.

1

LHS = $2\cos(x + \frac{\pi}{3})$
 $= 2(\cos x \cos \frac{\pi}{3} - \sin x \sin \frac{\pi}{3})$
 $= 2(\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x)$
 $= \cos x - \sqrt{3} \sin x = \text{RHS}$

(ii). Hence, or otherwise, solve the equation $\cos x - \sqrt{3} \sin x = \sqrt{3}$ for $0 \leq x \leq 2\pi$.

2

$\cos x - \sqrt{3} \sin x = \sqrt{3}$
 $2\cos(x + \frac{\pi}{3}) = \sqrt{3}$
 $\cos(x + \frac{\pi}{3}) = \frac{\sqrt{3}}{2}$
 $\cos^{-1}(\frac{\sqrt{3}}{2}) = \frac{\pi}{6}$
 $x + \frac{\pi}{3} = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}$
 $\therefore x = \frac{3\pi}{2}, \frac{11\pi}{6}$

$0 \leq x \leq 2\pi$
 $\downarrow$
 $\frac{\pi}{3} \leq x + \frac{\pi}{3} \leq \frac{7\pi}{3}$

9 Solve $\sqrt{3} \sin \theta + \cos \theta - \sqrt{3} = 0$ for $\theta \in [0, 2\pi]$.

3

$\sqrt{3} \sin \theta + \cos \theta - \sqrt{3} = 0$
 $\sqrt{3} \sin \theta + \cos \theta = \sqrt{3}$
 $\sqrt{3} \sin \theta + \cos \theta = r \sin(\theta + \alpha)$
 $= r \cos \alpha \sin \theta + r \sin \alpha \cos \theta$

Equating coefficients:

$r \cos \alpha = \sqrt{3}$
 $r \sin \alpha = 1$

$r^2 \cos^2 \alpha + r^2 \sin^2 \alpha = (\sqrt{3})^2 + 1^2$

$r^2 (\cos^2 \alpha + \sin^2 \alpha) = 4$

$r^2 = 4$

$r = 2$ (since $r > 0$)

and

$\frac{\sin \alpha}{\cos \alpha} = \frac{1}{\sqrt{3}}$

$\tan \alpha = \frac{1}{\sqrt{3}}$

$\alpha = \frac{\pi}{6}$

$2 \sin(\theta + \frac{\pi}{6}) = \sqrt{3}$

$\sin(\theta + \frac{\pi}{6}) = \frac{\sqrt{3}}{2}$

$\theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{13\pi}{3}, \dots$

$\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{25\pi}{6}, \dots$ but $0 \leq \theta \leq 2\pi$

$\therefore \theta = \frac{\pi}{6} \text{ or } \frac{\pi}{2}$